

Micro architecture

- Basics
- Microarchi
- state elements
- different Micro Archi
 - single cycle
 - multi cycle
 - pipelined
- Single cycle
- Pipelined.



Microarchitecture (Note)

→ Consider an example that, we designed a computer that adds two numbers and gave it our friend.

→ To use this computer, friend must know.

① what operation support (ADD)

② where to place operands (available Register)

③ where to get the result.

④ Rules need to follow (store data in Register
→ execute ADD instruction
→ result will be in the destination register specified in the instruction.)

These are informations are called architectural information

→ Computer architecture describes the rules and behaviour of a computer. It defines the instruction, register, data format, rules that tells the programmer what the computer does and what result to expect.

→ To implement a computer architecture, hardware is required

→ Same architecture can be implemented using different hardware designs.

→ Internal organization of this hardware is called micro-architecture

→ micro architecture is the internal design of the computer to implement architecture

→ micro architecture explains how the hardware is organized to make architectural rules to work.

Micro architecture has two parts

① Data path

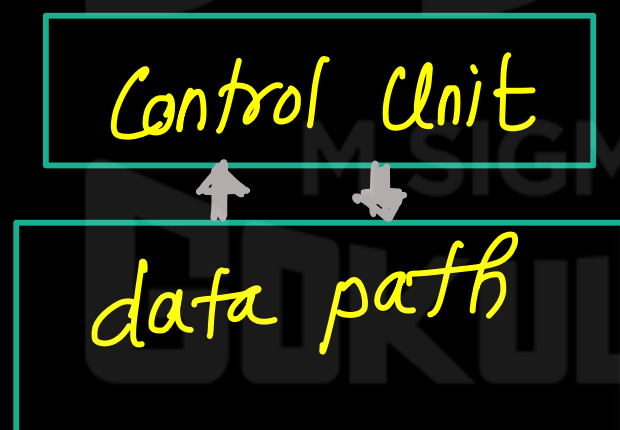
② Control unit

① Data path

- It is the hardware path way through which data flows inside the CPU
- Data path is the part of processor that contains the hardware used to store, move, and process the data during the execution of instructions.
- It contains structures such as memories, registers, ALU, multiplexers.

② Control unit

- Control unit receives the current instruction from the data path and tells the data-path how to execute the instruction
- Control unit produce different signal to control the operation of data path



State Elements in the data. (Note)

→ state elements are the hardware components to store and remember the information

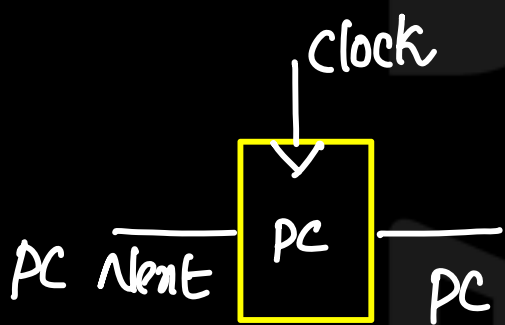
→ Different state elements are

- ① program counter (PC)
- ② Instruction memory (IM)
- ③ Register files
- ④ Data Memory.

Program Counter (PC):

— it is a register that store the address of next instruction to be executed.

→ In RISC processor instruction size is 4Byte (32bit)



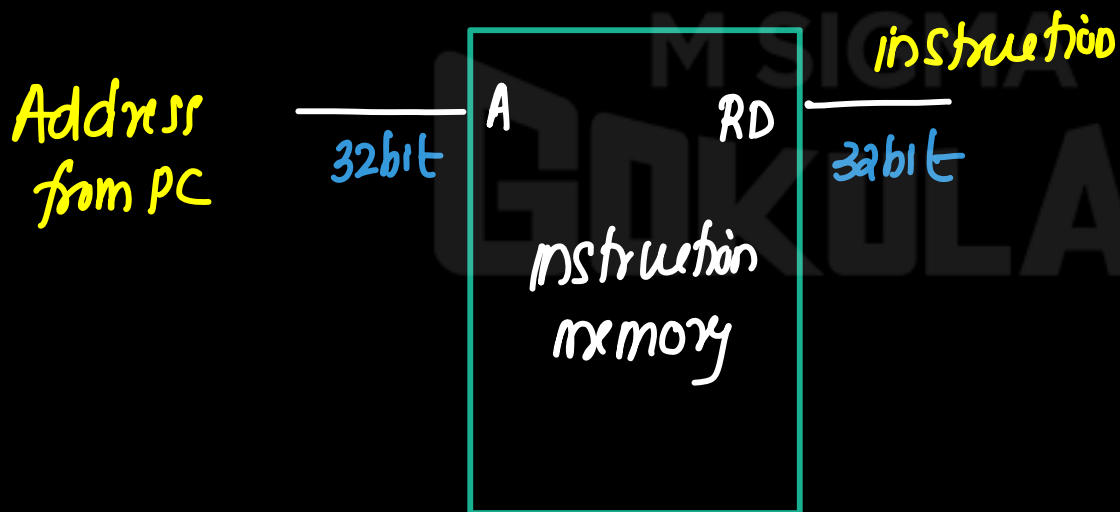
- PC Next → input to the program counter, indicates the address of next instruction to be executed
- PC → indicate the address of current instruction
- Address is 32 bit.
- clock is required to tell the pc when to update its val
- pc update its content at the edge of clock.



② Instruction Memory.

(N)

- this memory is used to store the instruction to be executed
- program counter provides the address and instruction memory outputs the instruction stored at that address. This instruction is sent to CPU for decoding and execution.
- Instruction memory has a single read port
- Read port is an interface through which instruction memory outputs an instruction when an address is given
- Read port has input lines through which 32 bit address is given to the instruction memory.
- Read port has output lines (read data output) (RD) through which memory outputs stored instruction.
- read port takes 32 bit address input (A) from PC and places the instruction on the read data output (RD)



Read data input - A
(address)

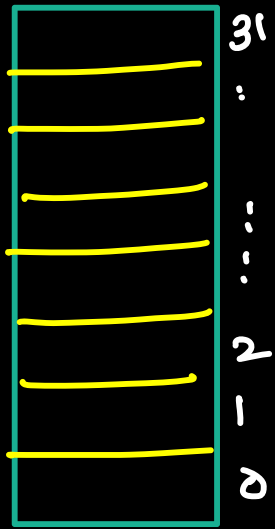
Read data output → instruction

③ Register file:

②

→ Register file is a collection of registers inside the CPU

→ Register file consists of 32 Register each of 32 bit size



$$32 \rightarrow 2^5$$

— so 5 bit Address is required to represent the register.

→ Address of Register $\Rightarrow$ 5 bit Address

→ two operations are possible in register file,

① Read : Read the content of Register through Read port

② Write : Write the data to the register through the write port.

→ Register files have two read port and one write port

→ Read port has

- ① read address input
- ② read data output

→ read address input takes the 5 bit address (specify the address of register read access) and register file load the content of register to register data output

read address input (A_1)
read data output (RD)

→ write port has

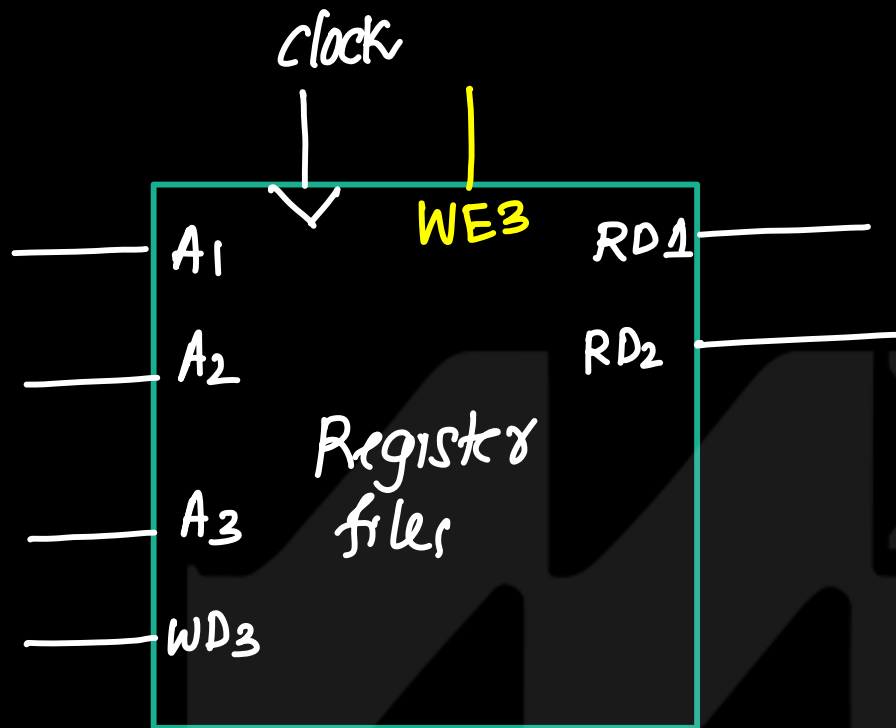
- ① write address input (A_3)
- ② write data (WD_3).

Register file store the data present at write data pin to the register whose address is specified in the write address inputs

→ Two operations can be performed in Registerfiles. These operations can be selected based on the values of control signal (write enable) (WE)

If $WE = 1$, perform write operation
If $WE = 0$, perform read operation

→ one clock input is required to control write operation. Data is written into the selected register only at the clock edges.



M SIGMA
GOKULAM

Data Memory

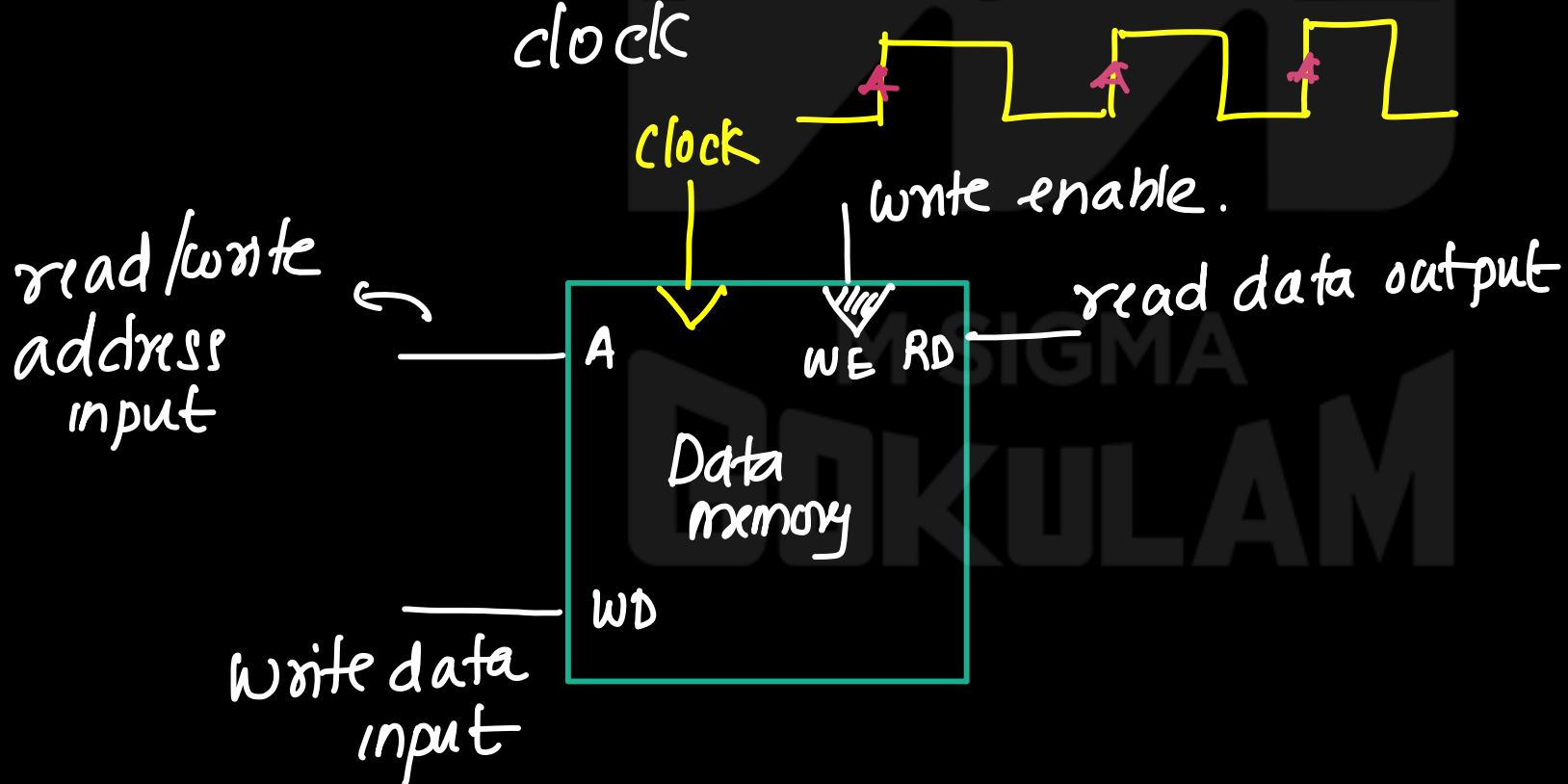
(N)

- This memory is used to store data
- This memory has single read write port
- Read write operation is selected on the value of write enable control signal

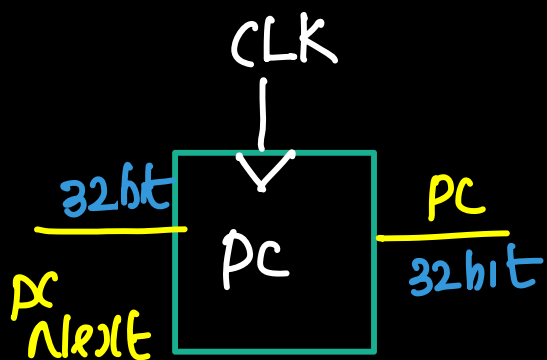
If write enable (WE) = 1, then data written on the location whose address present at the input line of read write port.

If $WE = 0$, then data is read from the memory location whose address is available at the input line of read/write port.

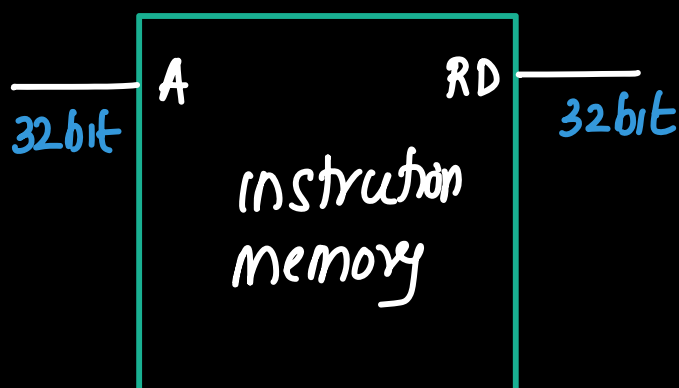
- A clock signal used to control the write operation write operation happens at the rising edge of clock



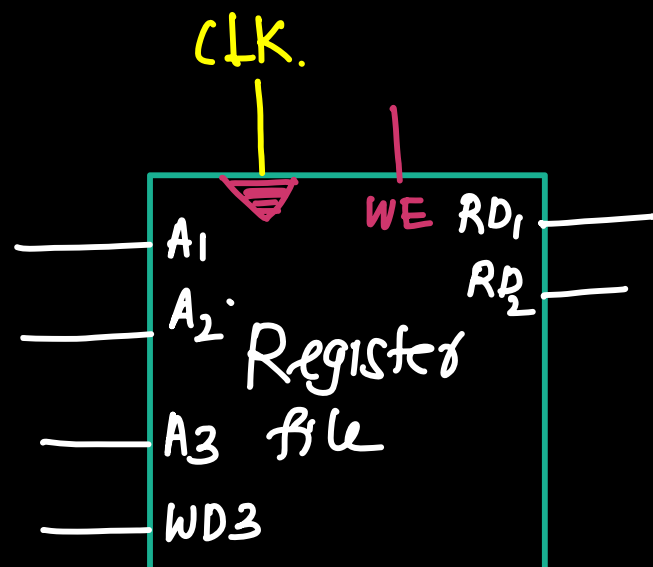
Summary



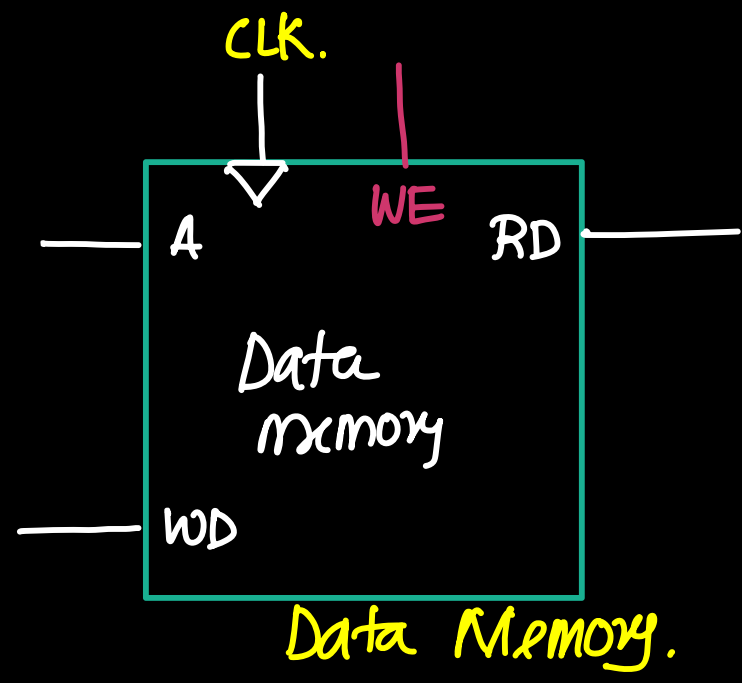
PC



Instruction memory (IM)



Register file



(N)



Commonly Used Microarchitecture (Note)

Single cycle microarchitecture:

- In this type, an instruction execute in one cycle
- What ever may be operation, one instruction execute in one cycle
- Let consider 3 instruction I_1, I_2, I_3 and T_1, T_2, T_3

are the time required to complete operation

$$T_1 > T_2 > T_3.$$

then cycle time is limited by slowest instruction.

(clock type $\leq T_i$)

- In this architecture, processor requires separate memory to store instruction and data.

Multicycle micro architecture

- In multicycle architecture, split the instruction execution into several stages. (IF, ID, execution, memory access, write back)
- each stage complete in one clock cycle.
- In this architecture instruction execute in a series of shorter cycles.
- Multicycle processor executes only one instruction at a time, but each instruction takes multiple clock cycle.

Pipelined Microarchitecture:

(N)

- In this architecture instructions are divided into several stages (IF, ID, EXE, memory access, write back) and multiple instructions are execute at same time.
- this architecture improves the processor performance by overlapping the execution of multiple instruction
- Each instruction require multiple clock cycle to complete.



Performance Analysis of Microarchitecture (Note)

→ Performance of microarchitecture refers to the how efficiently a processor execute the instruction. this is evaluated using execution time of a program.

→
$$\text{Execution time} = \text{no: of instructions} \times \text{CPI} \times \text{clock cycle time}$$

— no: of instruction = total no: of instructions in the program.

— CPI (clock cycle per instruction)
average no: of cycles required to execute one instruction

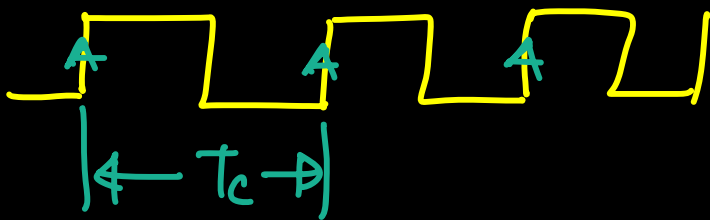
$$\text{CPI} = \frac{\text{no: of cycles}}{\text{Instruction}}$$

eg: $\text{CPI} = 1$ one cycle is required to complete one instruction.

$\text{CPI} = 4$ 4 cycle is required to complete one instruction

— clock cycle time (T_c)

time duration of one cycle.



→ $T_c = \frac{1}{f_c}$ $f_c \rightarrow$ no: of cycles in one sec

→ execution time of a program depends on

- ① no: of instruction
- ② CPI
- ③ T_c

- To reduce the execution time, reduce any one factor
→ But when we increase $f_c (\frac{1}{T_c})$, that doesn't provide better performance. So we need to design a good micro architecture to reduce the execution time

— throughput

No. of instructions completed in unit time

single cycle = low throughput

Multi cycle = Moderate throughput

pipelined = High throughput.



M SIGMA
GOKULAM